ARCR-UGV

300-WORD PROJECT SUMMARY • INSPIRE-MANAK AWARD

Project Overview

ARCR-UGV (Autonomous Range Cartridge Recovery and Hazard Mitigation Rover) is a proposed low-cost unmanned ground vehicle designed to make firing-range cleanup safer and more efficient. After training exercises, large numbers of spent brass cartridge cases can remain scattered across rough ground. Collecting them manually takes time and effort, and any suspected misfired or unexploded ammunition presents a serious hazard. The rover's central purpose is to reduce personnel exposure by supporting remote detection, recovery of confirmed spent casings, and collection of recyclable brass. The concept combines four main functions. First, an onboard camera and edge-computing system examine visible objects and flag items that may be hazardous. Computer vision cannot reliably prove that ammunition is safe, so any uncertain object must trigger a stop, distance-maintaining response, and alert to qualified explosive-ordnance personnel. Second, a metal-detection system is intended to locate metallic objects partly hidden beneath soil or gravel. Third, a controlled collection mechanism and conveyor guide suitable recovered casings into a sorting unit. A rotary sieve, or trommel, is proposed to separate casings by size into removable collection bins. Finally, tracked mobility and obstacle sensors support systematic coverage of uneven range terrain. ARCR-UGV aims to reduce repetitive manual work, improve range-clearance procedures, and recover brass for recycling. Its proposed performance targets—such as detection depth, cycle time, and manpower savings—must be verified through controlled testing rather than presented as proven results. The design should prioritize fail-safe behavior, remote operation, emergency stopping, clear status alerts, and documented testing. Suspected live ammunition must never be picked up, disturbed, or processed by the rover; it should be left for authorized specialists. The project's value lies in combining robotics, sensing, computer vision, and resource recovery around a practical safety problem. A staged prototype can begin with harmless metal objects and inert dummy casings in a controlled test area. Only after independent safety review and authorized validation should any operational deployment be considered. ARCR-UGV is therefore best presented as a safety-assistance and recycling concept, not an autonomous explosive-disposal system.